American Computer Science League

2022-2023 ● Contest 1: Next Base ● Intermediate Division

PROBLEM: Given 3 positive integers, *n*, *b*, and *s*, generate the next *n* numbers in base *b* starting with *s* in the given base. We guarantee that the base will be between 2 and 9 inclusive. We guarantee that *s* is a valid number in base *b*. Find the base 10 value for the number of times the largest possible digit in the given base is found among all of the digits in the numbers generated.

EXAMPLE: If *n*=15, *b*=8, and *s*=2, the numbers generated are 2, 3, 4, 5, 6, 7, 10, 11, 12, 13, 14, 15, 16, 17, 20. The largest possible digit in base 8 is 7 which occurs 2 times.

INPUT: There will be three integers representing the number of values to be found, the base to be used between 2 and 9 inclusive, and the starting value in the base given that will be no more than 16 digits.

OUTPUT: For each set of 3 input values, output a base 10 number representing the number of times the largest digit in the inputted base occurs in the sequence of numbers generated.

SAMPLE INPUT:

```
1. 15 8 2
2. 20 3 12
3. 25 5 324
4. 13 9 1652
5. 45 2 1111011
```

SAMPLE OUTPUT:

```
1. 2
2. 21
3. 24
4. 1
5. 170
```

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TEST DATA

TEST INPUT:

```
1. 1000 8 10
2. 50 4 13
3. 75 9 384
4. 100 6 555
5. 25 2 1100001110
```

TEST OUTPUT:

```
1. 357
2. 34
3. 13
4. 31
5. 121
```

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问题：给定 3 个正整数 n、b 和 s，生成以给定进制**的**数字 s 开始的接下来 n 个 b 进制数字。我们确保进制介于 2 和 9 之间（含 2 和 9）并且 s 是一个 b 进制的有效数字。请计算生成数字的各个数位上的所有数字**中，**在该进制下可能出现的最大数字一共出现的次数，以十进制表示。

示例：如果 $n = 15, b = 8$ 且 $s = 2$，生成**的数字是** 2、 3、 4、 5、 6、 7、10、11、12、13、14、15、 16、 17、20。八**进制**数各个数位上**最大**可能**数是** 7**，并且出现了** 2 **次。**

输入：将会有三个整数，分别表示生成值的数量，要使用的进制（介于 2 和 9 之间，包含 2 和 9），给定进制的起始值（不超过 16 位）。

输出：对于每组 3 个输入值，输出一个十进制数字，表示生成**的数列中输入进制**的数字各个数位上**的最大数字出现的次数。**

样本输入：

```
1. 15 8 2
2. 20 3 12
3. 25 5 324
4. 13 9 1652
5. 45 2 1111011
```

样本输出：

```
1. 2
2. 21
3. 24
4. 1
5. 170
```

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测试数据

测试输入：

```
1. 1000 8 10
2. 50 4 13
3. 75 9 384
4. 100 6 555
5. 25 2 1100001110
```

测试输出：

```
1. 357
2. 34
3. 13
4. 31
5. 121
```